QP Code 25900119

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCICH100 - MAJOR CHEMICAL INDUSTRIES

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple choice questions

Answer all questions

Each question carries 1 marks

1. Which type of glass is known for its high resistance to thermal shock? [K] / [CO1]
   a). Borosilicate glass (1 mark)

2. What is the main advantage of powder coating over traditional painting? [U] / [CO2]
   b). It is more environmentally friendly. (1 mark)

3. What is the main reason for the limited life cycle of secondary batteries? [U] / [CO2]
   c). Electrolyte depletion and electrode degradation. (1 mark)

4. In rechargeable batteries, which component undergoes structural changes over cycles? [U] / [CO3]
   b). Electrodes(1 mark)

5. In solar cell, photo current depends on.[U] / [CO3]
   d). All of these(1 mark)

6. Which of the following oxides has a direct bandgap, making it useful for optoelectronic applications?[A] / [CO1]
   b). Zinc oxide (ZnO) (1 mark)

7. Why are thermosetting plastics more suitable for applications requiring heat resistance?[U] / [CO4]
   c). They form permanent chemical bonds that prevent melting(1 mark)

8. Which type of gaseous fuel is produced by anaerobic digestion of organic waste?[U] / [CO1, CO5]
   c). Biogas(1 mark)

9. Which fraction is obtained first during the distillation of coal tar[U] / [CO1, CO5]
   b). Light oils(1 mark)

10. What is the primary purpose of the injection screw in injection moulding?[U] / [CO4]
    c). To melt and feed the plastic material into the mould(1 mark)

Part B
Short Answer Questions
Answer any 4 questions Each question carries 3 marks

11. Describe the characteristics and uses of traditional ceramics such as porcelain and earthenware.

Traditional Ceramics (Made from clay-based materials): Traditional ceramics have been in use for thousands of years and are mainly derived from natural raw materials like clay, feldspar, and silica. These ceramics undergo basic processes such as shaping, drying, and firing. Examples include pottery, porcelain, and tiles, which are widely used in construction, tableware, and decorative arts. These materials are valued for their durability, affordability, and resistance to environmental conditions. However, they typically exhibit lower mechanical strength and thermal resistance compared to advanced ceramics.
Examples: Bricks, tiles, pottery, porcelain
(Any relevant three points – 3 marks)

12. Discuss the composition and properties of armoured glass.

- Armoured glass, commonly known as bulletproof or ballistic glass, is designed to withstand high-impact forces while maintaining transparency.
- Its composition typically includes multiple layers of materials, each serving a specific purpose
- It is made up of layers of ordinary glass combined with polycarbonate sheets or other laminates.

  Composition:

  - Glass Layers – Toughened or laminated glass forms the outer layers for clarity and strength.
  - Polycarbonate or Polymer Interlayer – A flexible layer (often polyvinyl butyral or polycarbonate) helps absorb impact and distribute force.
  - Multiple Laminations – Several layers bonded together improve resistance to bullets, blunt force, or explosions.

- Advanced Composites – Modern variants may include ceramics or advanced polymers for increased protection.

Properties:

- Impact Resistance – Designed to absorb and disperse kinetic energy from bullets, projectiles, or impacts.
- Transparency – Despite its strength, it remains clear for visibility.
- Shatter Resistance – Fragments remain bonded due to interlayers, reducing risks of injury.
- Flexibility & Thickness Variation – Can be tailored to different security needs, ranging from basic protection to military-grade defense.
- Optical Clarity – Special coatings minimize distortion and glare.

Uses:

- Bank counters**,** military vehicles**,** bulletproof windows**,** and armored vehicles.

(Any relevant three points – 3 marks)

13. Write a note on the voltage generation in a battery.

A battery generates voltage through electrochemical reactions that occur in an electrochemical cell. Each cell consists of two electrodes—an anode and a cathode immersed in an electrolyte.At the anode, oxidation occurs, releasing electrons. These electrons flow through an external circuit to the cathode, where reduction takes place by accepting electrons. This movement of electrons constitutes an electric current, and the difference in electrical potential between the two electrodes is the voltage of the battery

Eg : A dry cell generates voltage through a chemical reaction between its components. It consists of a zinc outer casing (which acts as the anode) and a carbon rod surrounded by a manganese dioxide paste (which acts as the cathode). The electrolyte paste, typically ammonium chloride or zinc chloride, facilitates ion movement.When the circuit is connected, a chemical reaction occurs between the zinc and the electrolyte, releasing electrons at the anode. These electrons travel through the external circuit to the carbon cathode, creating an electric current. The potential difference between the anode and cathode determines the voltage, typically around 1.5V for a standard dry cell.

(Any relevant three points/ explanation of working of any specific cell – 3marks)

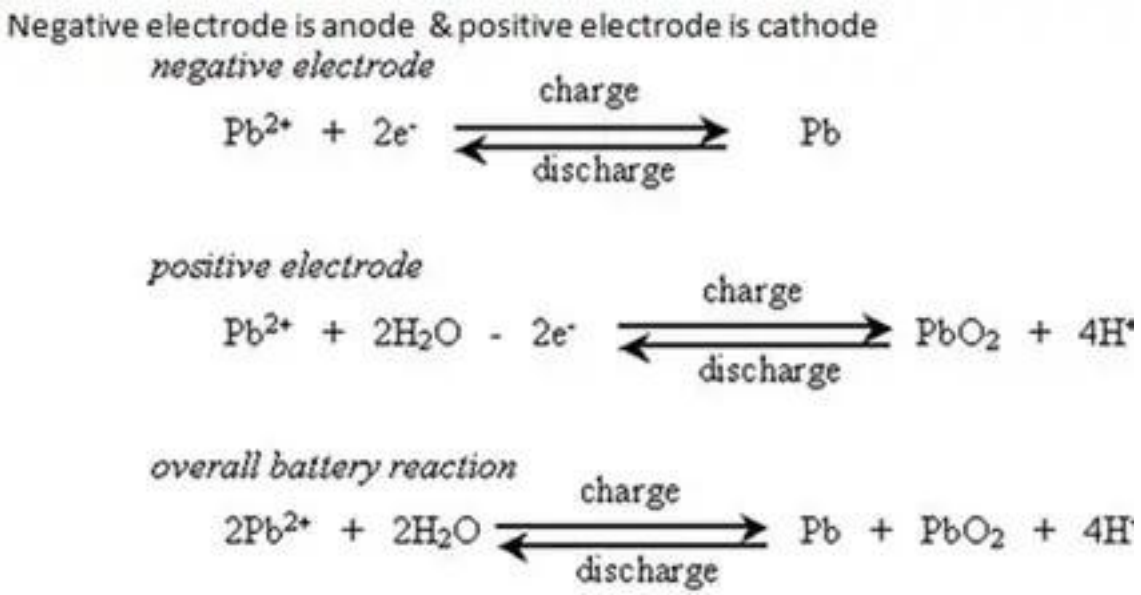
14. Write the anode and cathode reactions during the recharging of a lead acid battery.

- During Discharge:
  - The Pb anode (negative electrode) undergoes oxidation, releasing electrons.
  - The PbO_2 cathode (positive electrode) undergoes reduction, accepting electrons.
  - Sulfuric acid reacts with both electrodes, forming lead sulfate ($PbSO_4$) and releasing electrical energy.
- During Charging:

- An external current reverses the reaction, regenerating Pb and PbO_2 while breaking down $PbSO_4$

During the recharging process of a lead-acid battery, the reactions at the electrodes are essentially the reverse of the discharge reactions.

- **Anode (Negative Electrode)**: Lead sulfate ($PbSO_4$) is converted back into lead (Pb).
- **Cathode (Positive Electrode)**: Lead sulfate ($PbSO_4$) is converted back into lead dioxide (PbO_2).



(Correct reactions at electrode – 3 marks , Explanation of working of lead storage cell – 2 marks only)

15. Compare Transfer Moulding and Compression Moulding.

Transfer moulding is a process used to shape thermosetting materials by heating and pressing them into a mould. It begins with the material being placed into a chamber separate from the main cavity. When heat and pressure are applied, the softened material flows through channels into the mould, ensuring a precise and even distribution. This technique is particularly beneficial for producing complex shapes with embedded components, such as electronic parts. It also offers better control over material flow, reducing defects like air pockets, but tends to be more expensive due to additional equipment and material wastage.

Compression moulding, on the other hand, involves placing a pre-measured amount of material directly into an open mould cavity, then closing the mould and applying heat and pressure to shape it. This method is widely used for making strong, durable products like automotive parts and electrical insulators. Unlike transfer moulding, compression moulding is more cost-effective and requires simpler machinery, making it ideal for large-scale production. However, it can sometimes lead to variations in thickness and may not be suitable for intricate designs requiring fine details.

(Any three relevant points of both moulding – 3 marks)

16. Why is LNG preferred over CNG for long-distance transportation and heavy-duty applications?

LNG (Liquefied Natural Gas) is preferred over CNG (Compressed Natural Gas) for long-distance transportation and heavy-duty applications due to several key advantages,

- **Higher Energy Density**: LNG has a much higher energy density than CNG because it is stored in liquid form at extremely low temperatures. This means vehicles can travel longer distances on a single tank of LNG compared to CNG.
- **Lower Storage Volume**: Since LNG is denser, it requires less storage space than CNG for the same amount of energy, making it ideal for large trucks, ships, and industrial applications where space efficiency is critical.
- **Greater Range and Fuel Efficiency**: Heavy-duty trucks and transport vehicles benefit from LNG's extended range, reducing the need for frequent refueling stops, which improves operational efficiency.
- **Better Suitability for Heavy Loads**: LNG-powered engines generally provide better torque and power output than CNG engines, making them more suitable for hauling heavy loads over long distances.
- **More Practical Refueling Infrastructure**: LNG refueling stations are better suited for large-scale transportation fleets, while CNG is typically used for local and short-distance travel due to its refueling constraints.

LNG is a more efficient choice for long-haul and heavy-duty applications due to its superior storage, energy density, and performance characteristics.

(LNG and CNG definition -1 mark ,any two advantages – 2 marks)

Part C
Short Essay Questions
Answer any 4 questions Each question carries 7 mark

17. Analyze the role of carbon fibers in enhancing the mechanical properties of composite materials.

Carbon fiber is a strong, lightweight material composed of thin strands of carbon. These strands are woven into fabrics or embedded in resins to create composite materials.

Carbon fibers possess exceptional properties that make them highly valuable in various applications.

1. **High Tensile Strength** – Carbon fibers are incredibly strong, with a tensile strength far exceeding that of steel. This makes them ideal for reinforcement in composite materials.
2. **Low Density & Lightweight** – One of the most attractive features of carbon fibers is their low density, making them much lighter than metals while maintaining high strength.

3. **High Stiffness** – Carbon fibers exhibit excellent rigidity, ensuring minimal deformation under load. This is particularly useful in aerospace, automotive, and structural applications.
4. **Thermal Stability** – Carbon fibers maintain their mechanical properties at elevated temperatures, making them suitable for heat-resistant applications.
5. **Corrosion and Chemical Resistance** – Unlike metals, carbon fibers do not corrode and are resistant to chemicals, making them useful in harsh environments.
6. **Low Thermal Expansion** – Carbon fibers have a low coefficient of thermal expansion, meaning they do not expand or contract significantly with temperature changes.
7. **Electrical Conductivity** – Carbon fibers can conduct electricity, making them useful in applications such as electromagnetic shielding.
8. **Fatigue Resistance** – They have excellent resistance to fatigue and cyclic loading, maintaining their performance over extended use.

These properties enables the carbon fibres to enhance the mechanical properties of composite materials

(Carbon fiber definition – 2 marks, Any four properties- 5 marks)

18. Explain the role of pigments in paints, their classification, and their impact on paint properties.

Pigments are finely ground, insoluble particles that provide colour and opacity when mixed into a medium (e.g., paint, ink, or plastics).

Properties:

- Excellent color retention and stability.
- Insoluble in the medium, creating uniform dispersion.
- Resistant to light, heat, and chemicals.

Composition and Types:

1. **Organic Pigments**:
    - Derived from carbon-based compounds.
    - Example: Phthalocyanine blue for vibrant colors.
2. **Inorganic Pigments**:
    - Derived from mineral compounds.
    - Example: Iron oxide for earthy tones.
3. **Specialty Pigments**:
    - Include metallic and pearlescent pigments for unique visual effects.
    - Example: Aluminum flakes for metallic finishes.

Pigments play a crucial role in determining the properties of paints, influencing their color, durability, opacity, and even environmental impact. Here's how pigments affect the characteristics of paints:

1. **Color and Aesthetic Appeal** – Pigments provide the visual appeal of paints by reflecting and absorbing specific wavelengths of light. Their composition determines the vibrancy, depth, and shade of the final paint color.

2. **Opacity and Hiding Power** – The ability of a paint to cover the surface beneath it depends on the opacity of its pigments. High-opacity pigments improve coverage, reducing the number of coats needed to achieve the desired effect.
3. **Durability and Weather Resistance** – Some pigments enhance the resistance of paints to UV radiation, moisture, and temperature fluctuations, preventing fading and degradation over time. This is especially important in exterior paints.
4. **Chemical and Corrosion Resistance** – Pigments can improve paint's ability to resist chemical exposure, making them valuable in industrial coatings for metal surfaces and harsh environments.
5. **Reflectivity and Heat Absorption** – Certain pigments influence how much heat a surface absorbs. Light-reflective pigments help reduce heat buildup in buildings and vehicles, contributing to energy efficiency.
6. **Environmental and Health Considerations** – Traditional lead-based pigments were phased out due to toxicity concerns. Modern paints use eco-friendly and non-toxic pigments to improve safety for humans and the environment.
7. **Special Effects and Functionality** – Some pigments are designed for specific purposes, such as luminescent or fluorescent pigments for safety signage, or metallic pigments for decorative finishes.

   (Definition and properties – 2 marks, Classification – 3 marks, Impact on paint properties- 2 mark)

19. Describe the function and importance of a salt bridge in an electrochemical cell.

Salt bridge is usually an inverted U-tube filled with concentrated solution of inert electrolytesGenerally salts like KCl, KNO3, and K2SO4 etc. are used. For the preparation of salt bridge, gelatin or agar-agar is dissolved in a hot concentrated aqueous solution of an inert electrolyte and solution thus formed is filled in the U-tube. On cooling the solution set in the form of a gel in the U-tube. The ends of the U-tube are plugged with cotton wool as to minimize diffusion effects.

Function of salt bridge: (a) It connects the solutions of two half cells and complete the cell circuit. (b) It prevents transference or diffusion of the solutions from one half cell to the other. (c) It keeps the solutions in the two half-cells, electrically neutral.

- **Prevents Charge Buildup** – In an electrochemical cell, oxidation occurs in the anode half-cell, releasing positive ions into the solution, while reduction takes place at the cathode, consuming positive ions. Without a salt bridge, an excess of charge would develop, disrupting the reaction.

- **Completes the Circuit** – The salt bridge provides a pathway for the movement of ions, allowing the reaction to continue by enabling the flow of electric current between the electrodes.

- **Maintains Ion Balance** – It contains an inert electrolyte (usually KCl or KNO_3), which allows the free movement of cations and anions to counteract charge imbalances while preventing unwanted side reactions.

- **Prevents Direct Mixing** – Unlike a porous partition, which physically separates the solutions, the salt bridge keeps them from mixing directly while still allowing controlled ion exchange.

(Definition with any two relevant points – 3 marks, functions – 4marks)

20. How does the Nernst equation explain the variation of electrode potential with concentration?

The Nernst equation is a crucial tool in electrochemistry, allowing for the calculation of electrode potential and cell potential under non-standard conditions

Consider the reduction reaction taking place at electrode when it is immersed in its salt solution

$M^{n+} + ne^- \longrightarrow M$

The Nernst equation for the electrode potential of the electrode is given as

$E_{Mn+/M} = E^o_{Mn+/M} - \frac{RT}{nF} \ln Q$

Where Q is the reaction quotient $= \frac{[Prouct]}{[Reactant]} = \frac{[M]}{[Mn+]}$

$E_{M^{n+}/M} = E^0_{M^{n+}/M} - \frac{RT}{nF} \ln \frac{[M]}{[M^{n+}]}$

For pure metals, [M]=1

$E_{M^{n+}/M} = E^0_{M^{n+}/M} - \frac{RT}{nF} \ln \frac{1}{[M^{n+}]}$

$E_{M^{n+}/M} = E^0_{M^{n+}/M} - \frac{2.303RT}{nF} \log \frac{1}{[M^{n+}]}$

At 25°C, substituting R= 8.314JK^{-1}mol^{-1}

F= 96500 C

T= 298K

$\frac{2.303RT}{nF} = \frac{0.0591}{n}$

$E_{M^{n+}/M} = E^0_{M^{n+}/M} - \frac{0.0591}{n} \log \frac{1}{[M^{n+}]}$

(Nernst equation – 3 marks, Explanation of electrode potential/derivation – 4 marks)

21. Describe the Process, Types, and Applications of Foaming in Plastics.

Process of Foaming:

The foaming process involves incorporating gases (like air, nitrogen, or vapors from low-boiling solvents such as *n*-pentane) into a polymer matrix. Surface-active agents aid the formation of a cellular structure. The polymer is then cooled or cured to stabilize the foam.Foaming in plastics is a process that creates a cellular structure within the material, significantly reducing its density while enhancing its insulating properties, flexibility, and strength-to-weight ratio.This can be achieved through various methods:

1. **Physical Foaming** – Gases like CO_2 or nitrogen are injected into molten plastic, expanding upon cooling to create foam.
2. **Chemical Foaming** – Chemical blowing agents decompose at elevated temperatures, releasing gases that cause the polymer to foam.
3. **Mechanical Foaming** – Air or gas is mechanically whipped into liquid polymer before solidifying into foam.

4. **Extrusion and Molding Techniques** – Plastics can be foamed via extrusion or molding processes, controlling factors like temperature and pressure to achieve specific properties.

Types of Foamed Plastics

Foamed plastics are classified based on their structure and applications:

- **Rigid Foams** – High-strength, stiff materials used for insulation and structural components (e.g., polyurethane, polystyrene).
- **Flexible Foams** – Soft, compressible foams used in cushioning and packaging (e.g., polyethylene, polyurethane).
- **Open-Cell Foams** – Porous structure with interconnected cells allowing air and liquid flow (common in sound insulation).
- **Closed-Cell Foams** – Gas-filled, sealed cells providing superior insulation and buoyancy (used in flotation devices and thermal insulation).

Applications of Foamed Plastics

- **Construction** – Thermal and acoustic insulation panels, sealants, and structural components.
- **Automotive** – Lightweight interior parts, soundproofing, and impact-absorbing materials.
- **Packaging** – Shock-absorbing foam for fragile items and food containers.
- **Medical** – Prosthetics, cushioning materials, and protective padding.
- **Sports and Leisure** – Foam-based equipment like yoga mats, helmets, and flotation devices.
- **Consumer Products** – Mattresses, furniture cushions, and cleaning sponges.

(Process – 3 marks, Types – 2 marks and Applications - 2 marks)

22. Describe the ultimate analysis of coal and how it differs from proximate analysis.

Ultimate analysis determines the *elemental composition* of coal, sit involves the eterminthe percentage of carbon (C), hydrogen (H), nitrogen (N), sulfur (S), and oxygen (O) , Ash and Moisture present.. It helps evaluate the chemical characteristics of coal, which are important for combustion calculations, environmental impact assessment (e.g., sulfur emissions), and fuel value estimation.The analysis is done through chemical reactions and combustion techniques in laboratory settings using instruments like CHNS analyzers.

Proximate analysis of coal is a standardized method used to assess its composition and suitability for different applications. It provides key insights into the coal's physical and chemical properties by determining the following parameters:

1. **Moisture Content** – The amount of water present in coal, which affects its calorific value. High moisture reduces the efficiency of combustion and transportation.
2. **Volatile Matter** – The fraction of coal that is released as gas or vapor when heated in the absence of air. It influences the coal's ignition characteristics and combustion behavior.

3. **Ash Content** – The inorganic residue left after combustion, composed mainly of minerals. Excessive ash lowers the heating value and contributes to slag formation in boilers.
4. **Fixed Carbon** – The solid combustible portion remaining after volatile matter is driven off. It determines the coal's ability to generate heat and is crucial for applications like metallurgical coke production.

Key Differences Between Ultimate and Proximate Analysis:

1. Ultimate Analysis measures the elemental composition of coal (C, H, S, N, O). Proximate Analysis focuses on physical characteristics rather than chemical composition.
2. Ultimate Analysis helps assess the fuel quality, emission characteristics, and combustion behaviour. Proximate Analysis determines moisture content, volatile matter, fixed carbon, and ash percentage
3. Ultimate Analysis is essential for environmental studies, especially regarding sulfur emissions and pollutants.Proximate Analysis is useful for understanding coal's immediate combustibility and industrial applications.
4. Proximate analysis is a quick assessment used for classification and usage decisions, ultimate analysis provides deeper insights into the coal's chemical nature, aiding in precise energy calculations and pollution control strategies
5. Ultimate analysis is used for chemical assessment and combustion calculations where as Proximate analysis is used for quick classification and fuel handling decisions.
6. Ultimate analysis requires sophisticated instrumentation (e.g., CHNS analyzers).where as Proximate analysis uses simpler lab methods like heating and weighing.

(Descrption of ultimate analysis – 3 marks , Proximate analysis – 2 marks ,
Any two difference -2 marks)